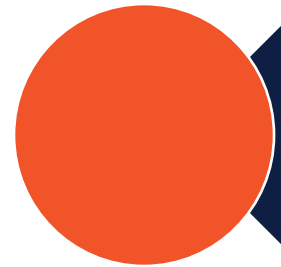


Introduction to Statistics

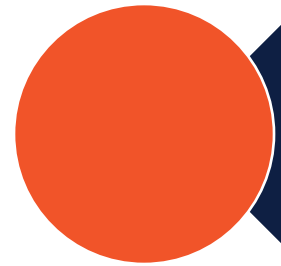
Team Teaching Computational Statistics

2024/2025

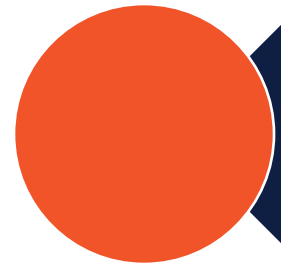
Outline



Intro to Statistics



Variables



Data

Introduction to Statistics

They way of reasoning

Think about this

DEDUCTION



Doctors perform medical actions and diagnoses based on patient data, research, and prior knowledge.

INDUCTION



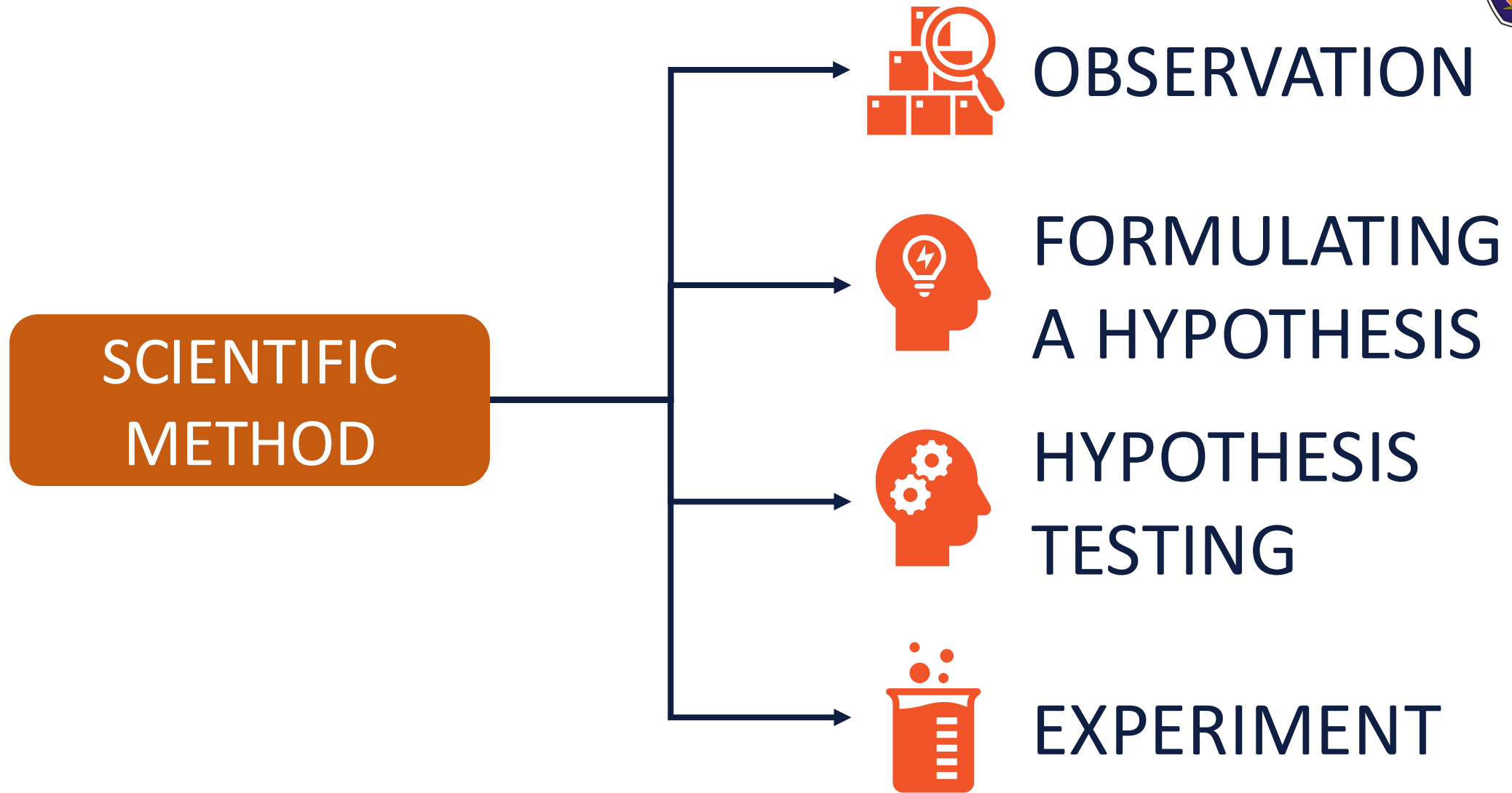
Neurology researchers study brainwave activity from several subjects and attempt to draw conclusions about common phenomena.

SCIENTIFIC METHOD



Carrying out scientific steps to prove a hypothesis about a phenomenon through inductive inference.

Scientific Method



Scientific Method - Example



The research was conducted on 20 patients with a specific disease. To draw general conclusions based on the available data, the researchers might have questions such as:

1. If 20 new patients are studied, will the results be the same as those from the previous 20 patients?
2. If different labs analyze the same blood sample, will the results be the same?
3. If the blood sample is stored at different temperatures, will the results be the same?

So?

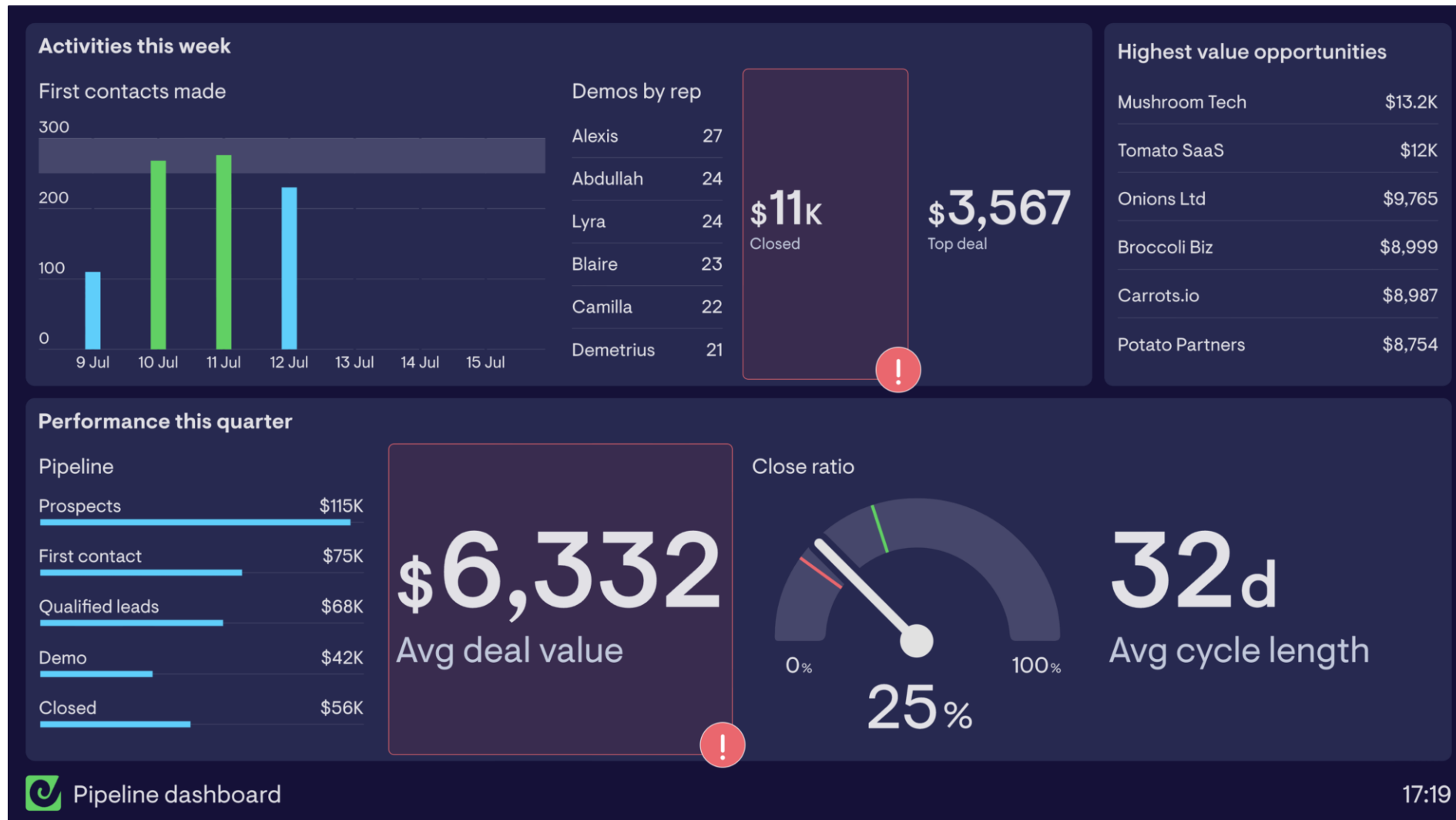
How to prove or test our hypothesis?

Sample → Population , Statistical Test, Draw a Conclusion

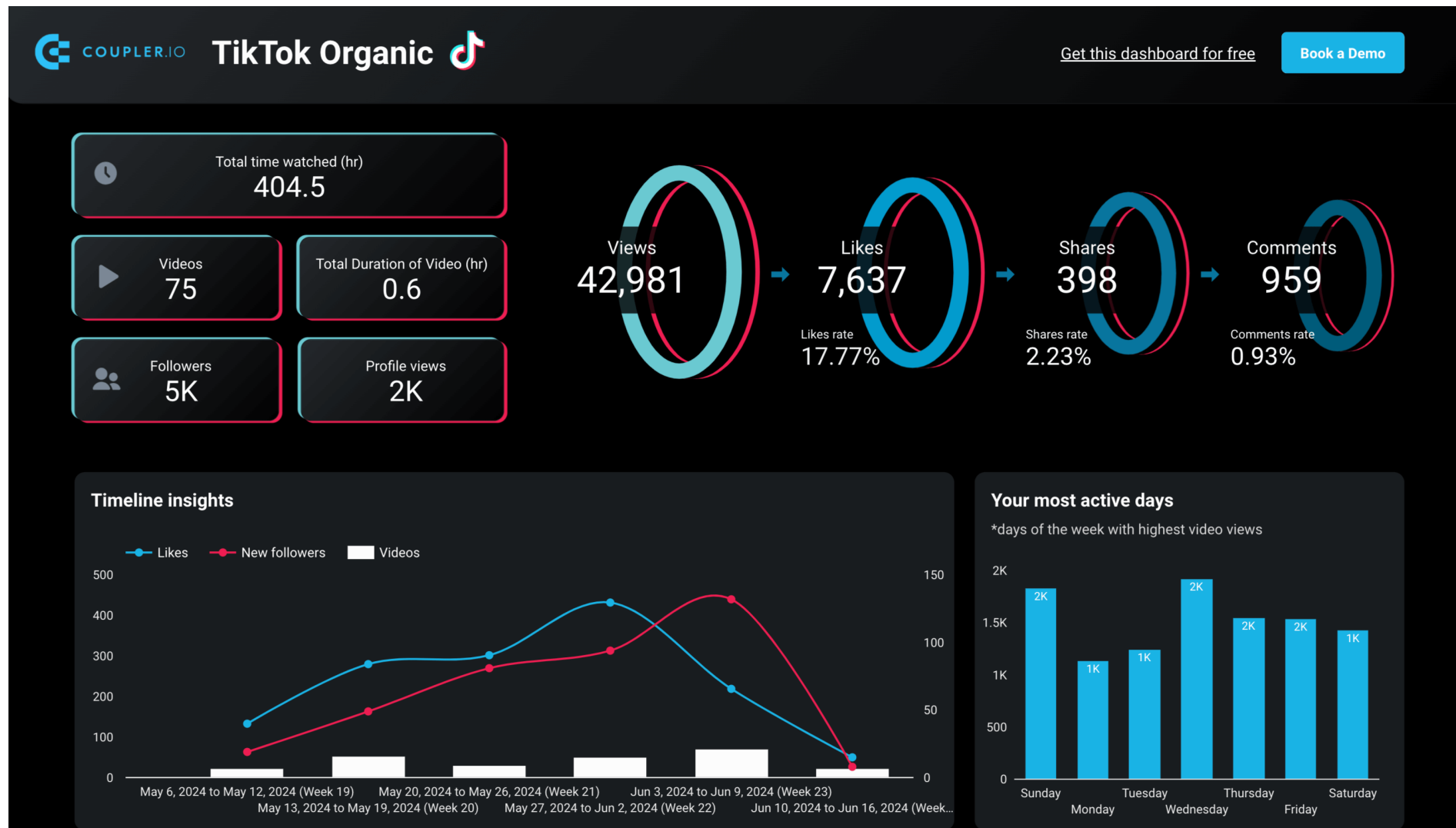
Introduction to Statistics

We encounter statistics every day

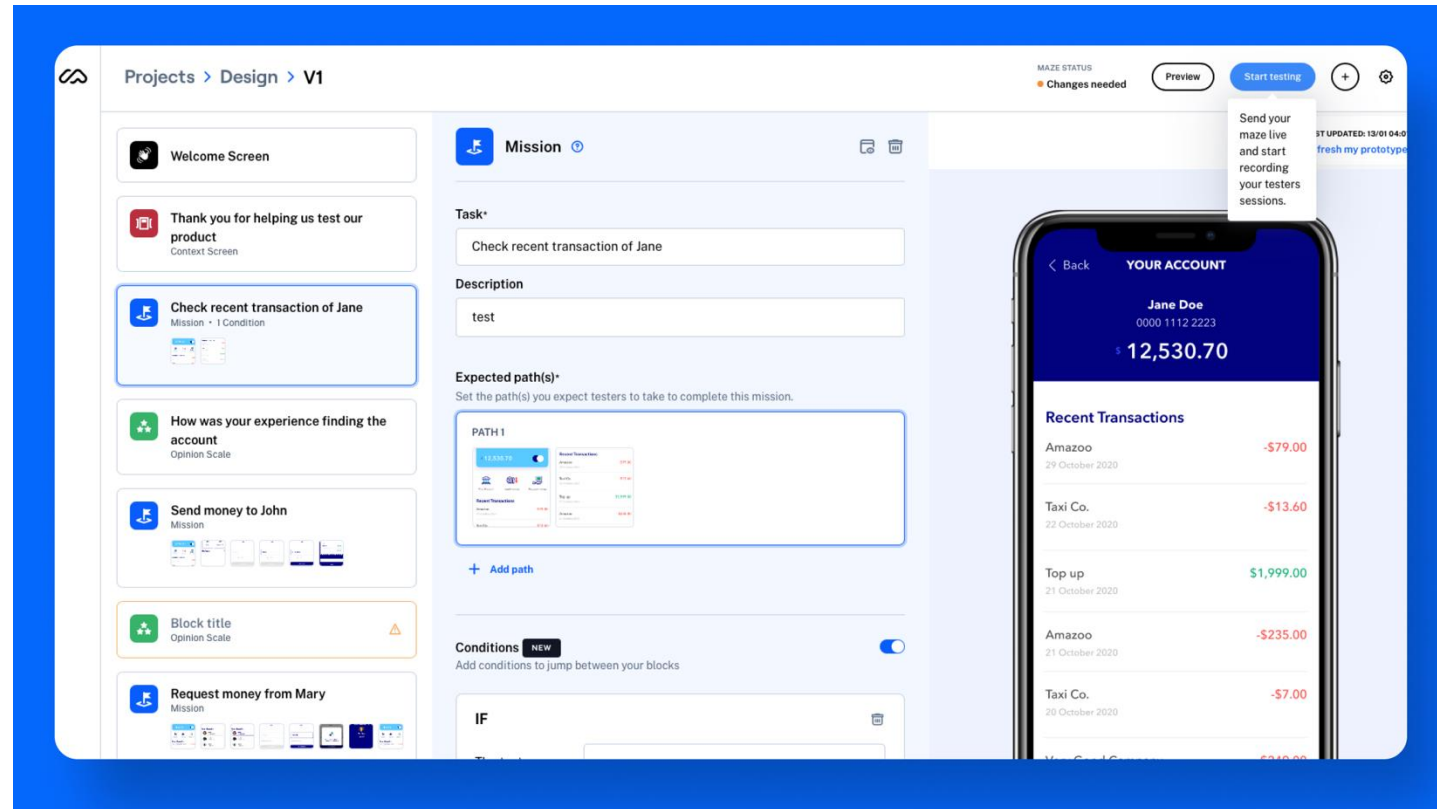
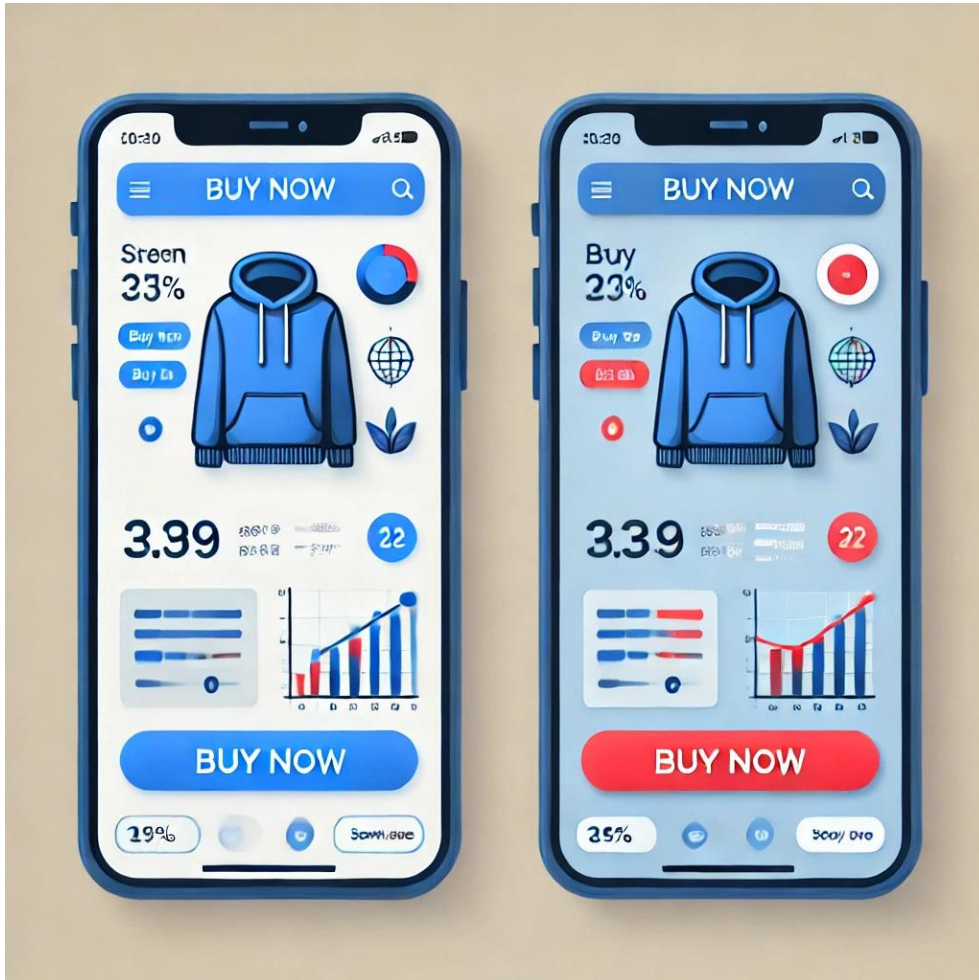
Sales Dashboard



Social Media Engagement



A/B Testing UI/UX in Apps



So, what is **statistics**?

What is statistic? What is statistics?

What is Statistics?

Statistic



The estimation of unknown numerical quantities, such as the mean, median, or mode. (Sumarjaya, 2016)

Statistics



The discipline concerned with the collection, organization, analysis, interpretation, and presentation of information that can be expressed numerically. (Sumarjaya, 2016)

Statistics are procedures that combine, organize, and summarize data to make inferences (Burnham, 2015)

When, where, and why

When do we use statistics?

- Whenever we seek answers to a phenomenon that requires a summary of data.

Where can we find statistics?

- In almost every aspect of modern life: business, research, education, government, the military, and sports.

Why do we need statistics?

- Research → To understand phenomena based on data findings
- Explaining phenomena → Used to describe a phenomenon more concisely and in a way that is easier to understand (upper-level explanation)

Types of Statistics



Descriptive Statistics

Consolidating (combining) data into a single value that represents the entire dataset.



Inferential Statistics

Analyzing the relationships between variables studied in a phenomenon

Making inferences and assumptions about the collected data to draw conclusions about a phenomenon

Statistics

Variables

What will be measured, investigated, or concluded in a study?

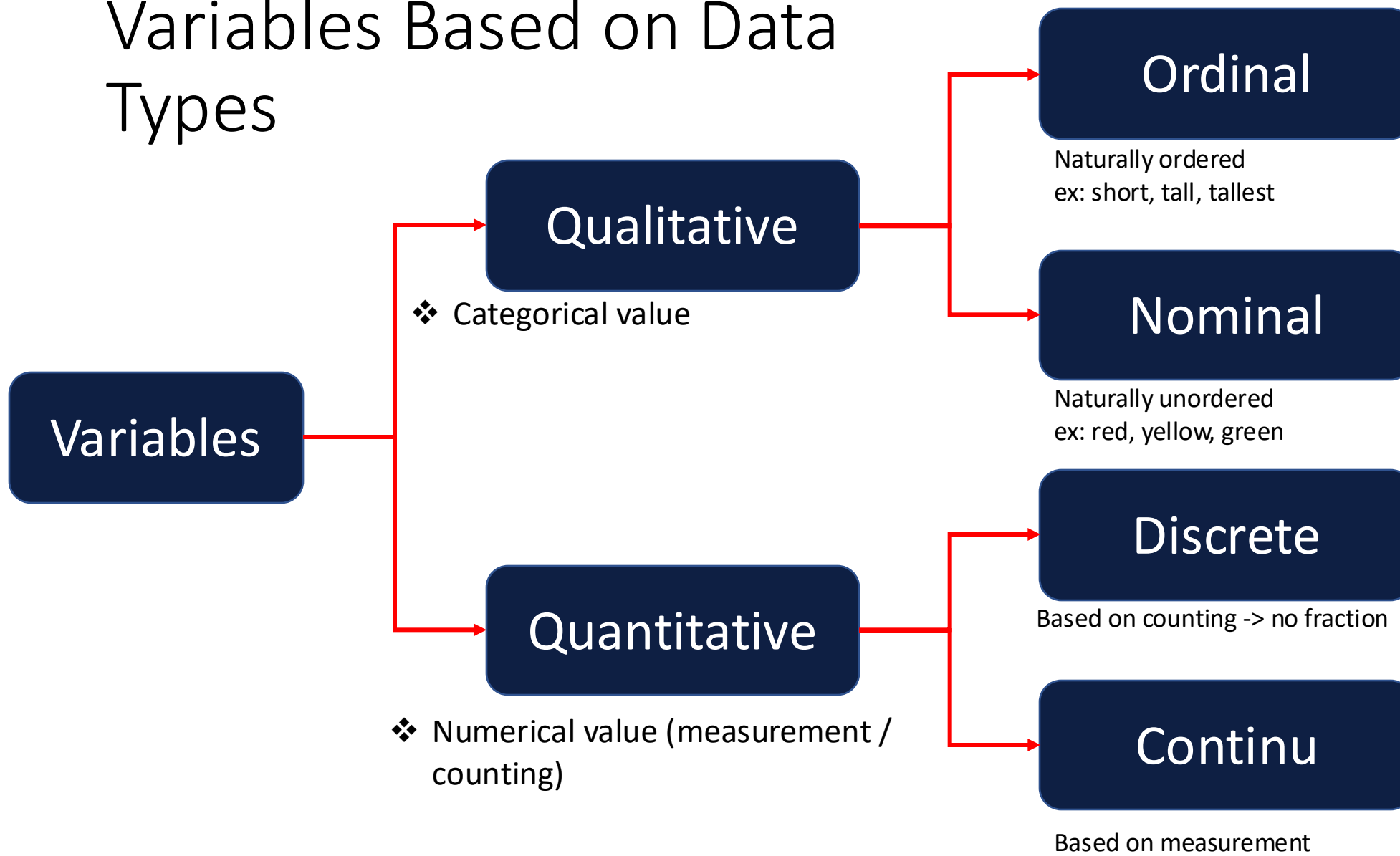
What is variables?



Everything that will be the object of observation in the research or the factors involved in the events or phenomena being studied.
(Suryabrata, 1983)

A concept or characteristic that has varying values.
(Kerlinger, 2004)

Variables Based on Data Types



Unclear point
Agreed range



Clear point

Qualitative: Ordinal Variables



How do you feel today?

- ☒ 1 – Very Unhappy
- ☐ 2 – Unhappy
- ☐ 3 – OK
- ☐ 4 – Happy
- ☐ 5 – Very Happy

How satisfied are you with our service?

- ☒ 1 – Very Unsatisfied
- ☐ 2 – Somewhat Unsatisfied
- ☐ 3 – Neutral
- ☐ 4 – Somewhat Satisfied
- ☐ 5 – Very Satisfied

Sumber: <https://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio/>

Qualitative: Nominal Variables



What is your gender?

- ☒ M - Male
- ☐ F - Female

What is your hair color?

- ☒ 1 - Brown
- ☐ 2 - Black
- ☐ 3 - Blonde
- ☐ 4 - Gray
- ☐ 5 - Other

Where do you live?

- ☒ A - North of the equator
- ☐ B - South of the equator
- ☐ C - Neither: In the international space station

Sumber: <https://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio/>



How to quantify the qualitative data?

It should be can be calculated by using mathematical formula

Converting to Quantitative Measurement

How to do that?

ENCODING

- Convert it into a number
- Ex: Male → 1; Female → 0
- There are various method for encoding

Scaling

- Turn into scaling value → *Likert scale*

How satisfied are you with *

	Very Unsatisfied	Unsatisfied	Neutral	Satisfied	Very Satisfied
Purchase	☐	☐	☐	☐	☒
Service	☐	☐	☐	☒	☐
Company Overall	☐	☐	☐	☐	☒

Variables Based on Its Function #1



- Also known as an independent variable or predictor
- Actively controlled by the researcher to understand changes in the dependent variable

- Also known as the dependent variable, response, or outcome
- Influenced by the conditions of the independent variable
- Represents the primary objective of the research

FEATURES



PREDICTION

Paradigm in machine learning

Variables Based on Its Function #2

Moderator

- A variable that influences the dependent variable but is not the primary focus.

Intervening

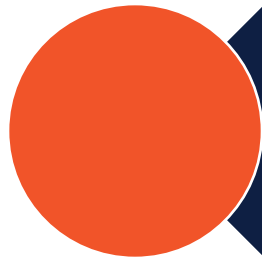
- A variable that, in theory, influences the relationship between the dependent and independent variables. It can strengthen or weaken the relationship between the dependent and independent variables.

Control

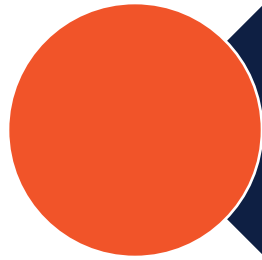
- A factor that is controlled when comparing the effects of other variables.

Then, what the different between variable and data?

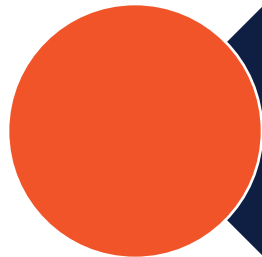
What is data?



The actual value of the variable
(Illowsky, dkk., 2022)



Unit of information



Need to be transformed into
information

Data vs. Variables



Variable

How do you feel today?

- ☒ 1 – Very Unhappy
- ☐ 2 – Unhappy
- ☐ 3 – OK
- ☐ 4 – Happy
- ☐ 5 – Very Happy

How satisfied are you with our service?

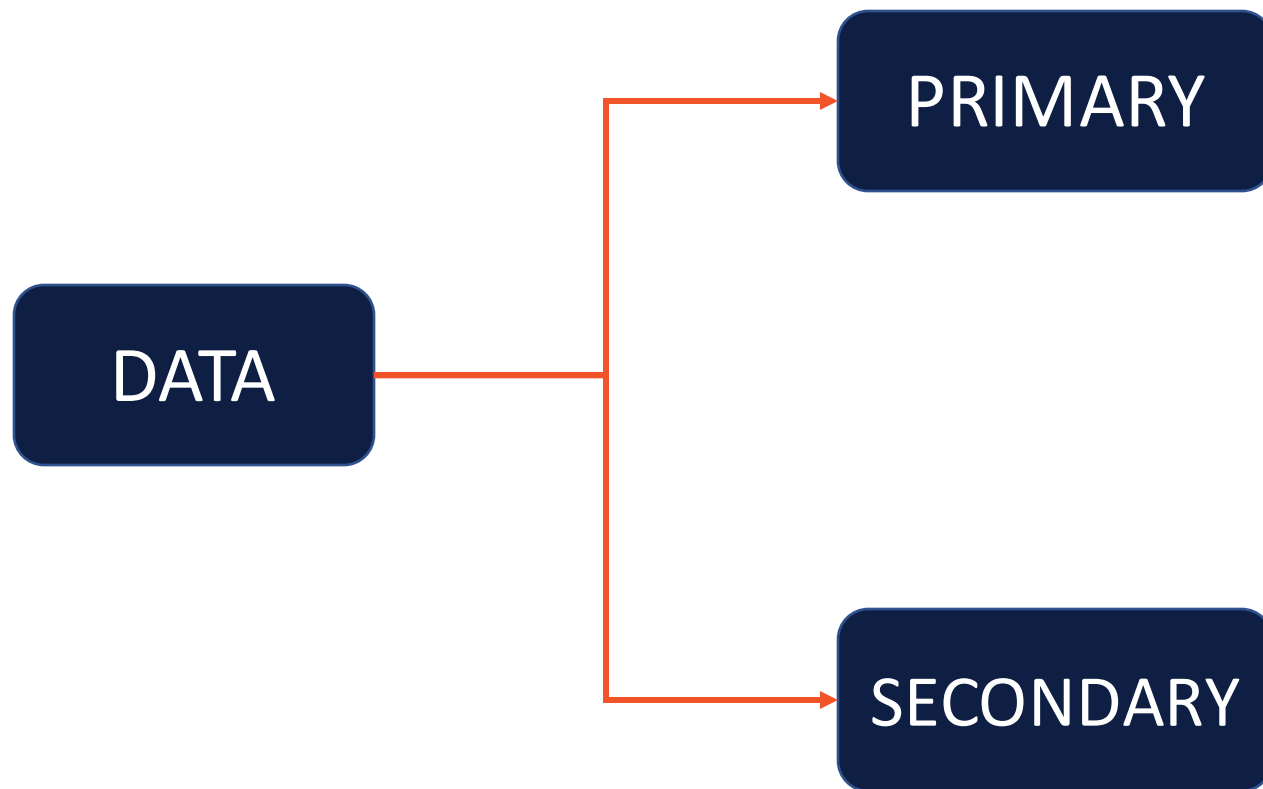
- ☒ 1 – Very Unsatisfied
- ☐ 2 – Somewhat Unsatisfied
- ☐ 3 – Neutral
- ☐ 4 – Somewhat Satisfied
- ☐ 5 – Very Satisfied

Data

Data vs. Variable – Another Example

Voluntariness of use	1	2	3	4	5	6	7
My participation on Facebook during the demonstration were voluntarily							
Constructs	1	2	3	4	5	6	7
Performance expectancy							
PE01: I would find Facebook useful by providing the news at the moment about the demonstration and what is going on right now. (Increasing of awareness)							
PE02: Through using Facebook can I publish demonstration events to the world more quickly than traditional media							
PE03: Through using digital media such Facebook, my physical participations will increase (Facebook motivate me to go down to street and contribute in demonstration)							
Effort expectancy	1	2	3	4	5	6	7
EE01: My comments which I have written about demonstration more clear and expression while it organized by one page "Khalid Saied Page" which concern one subject							
EE02: It would be easy to know which pages concern about demonstration events.							
EE03: I would find Facebook easy to use and flexible and no restrictions to join or leaving the page.							
Social influence	1	2	3	4	5	6	7
SI01: participants in demonstration who have influence (relatives) my behavior think that I should use Facebook.							
SI02: participants in demonstration who are important to me (non- relatives) think that I should use Facebook.							
SI03: In general, the community encourages to use Facebook and share the information as much as can							
Facilitating conditions	1	2	3	4	5	6	7
FC01: I have necessary resources to use Facebook. (The government allows and control the internet traffic during the demonstration)							
FC02: I have necessary knowledge to use Facebook.							
FC03: The Facebook is not the only method which deals with during the demonstration							
FC04: I do not need assistance to deal with Facebook functions.							
Behavioral intention to use the system	1	2	3	4	5	6	7
BI01: I will work on publishing demonstration news on Facebook pages.							
BI02: I will continue to publish demonstration events without stop							

How to gather the data?



- Collected by researcher
- Its validity remaining in question

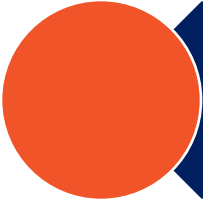
- Collected by another researcher
- Commonly valid (proved by community)

How to collect primary data

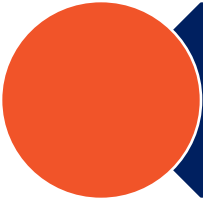


-  Registration
-  Survey
-  Census
-  Observation

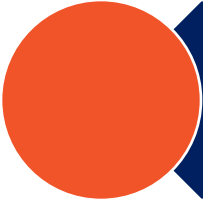
How to collect secondary data



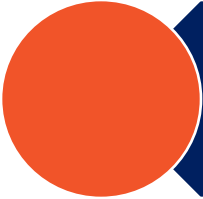
Scientific Publication



Data from Institution (Gov/Private)



Data Portal



etc

Learning Tools



Post-test



- Classify the following data into the appropriate type: quantitative (discrete/continuous) or qualitative (ordinal/nominal):
 - Blood type: O, A, B, AB
 - Gender: Male/Female
 - Blood pressure, height, weight, age
 - Number of children in a class
 - Disagree, neutral, agree

